Assignment 2: Key Concepts and Problem Map

Chapter 2 — matrix algebra, inverses, rank, LU

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The problem map: 17 questions, six ideas

Concept cluster	Problems	The question it asks	Deciding test
A. Products & shapes	Q1 Q15 Q16	Is AB defined, and is it BA?	inner dimensions match; $AB \neq BA$ in general
B. Invertibility	Q2 Q3	does A^{-1} exist, and what is it?	$\det \neq 0$; 2×2 formula
C. Matrix equations	Q4 Q5 Q6	isolate X, respecting the side	$X = A^{-1}B$; order reverses for products
D. Rank & nullity	Q7 Q8 Q9	how many free variables?	Rank Theorem: $\text{rank} + \dim \text{Nul} = n$
E. LU factorization	Q10–Q14	factor $A = LU$, then use it	row additions recorded in L
F. Elementary matrices	Q17	row op as a matrix	perform the op on I

- Within a cluster the problems are the *same question with different arithmetic*: solve one carefully, the rest follow.

A · Products and shapes (Q1, Q15, Q16)

Q1 — compute AB, BA; then a T/F

$A = \left[\begin{smallmatrix} -1 & 1 \\ 4 & 1 \end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix} 3 & 2 \\ 2 & -3 \end{smallmatrix}\right]$:

$$AB = \begin{bmatrix} -1\cdot 3+1\cdot 2 & -1\cdot 2+1\cdot(-3) \\ 4\cdot 3+1\cdot 2 & 4\cdot 2+1\cdot(-3) \end{bmatrix} = \left[\begin{smallmatrix} -1 & -5 \\ 14 & 5 \end{smallmatrix}\right], \quad BA = \left[\begin{smallmatrix} 5 & 5 \\ -14 & -1 \end{smallmatrix}\right] \Rightarrow AB \neq BA \Rightarrow \textbf{False}.$$

Q15 — one matrix, two products

$A = (-2, 3, 4)^\top$ (a column):
$A^\top A = (-2)^2 + 3^2 + 4^2 = \mathbf{29}$ (a 1×1 — enter a number); $AA^\top$ is the full 3×3 $[[4, -6, -8], [-6, 9, 12], [-8, 12, 16]]$.

Q16 — shapes decide

A, B: 4×9; C: 3×4. Defined: **A** $AB^\top$ $(4\times9)(9\times4)$, **D** $A^\top$ (9×4), **E** $B - A$ (same shape), **F** CA $(3\times4)(4\times9)$. Not defined: AC $(9 \neq 3)$, $C + A$ (shapes differ).

B · The 2×2 inverse test (Q2, Q3)

Q2 — does the inverse exist?

$M = \left[\begin{smallmatrix} 2 & 3 \\ 3 & 4 \end{smallmatrix}\right]$: $\det M = 2{\cdot}4 - 3{\cdot}3 = -1 \neq 0 \Rightarrow$ **exists**.

$$M^{-1} = \tfrac{1}{-1}\left[\begin{smallmatrix} 4 & -3 \\ -3 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -4 & 3 \\ 3 & -2 \end{smallmatrix}\right]$$

Check: $MM^{-1} = I$ ✓

Q3 — invertible for which c?

$A = \left[\begin{smallmatrix} 1 & 1 \\ c & c^2 \end{smallmatrix}\right]$:

$\det A = c^2 - c = c(c-1) \neq 0 \;\Rightarrow\; c \neq 0$ and $c \neq 1$.

One test, everywhere

$\det \neq 0$ is the Chapter 2 way to say "invertible" — the same fact the IMT states via pivots, rank, and trivial null space.

C · Matrix equations: isolate X without touching order (Q4, Q5, Q6)

Q5 — the warm-up

$AX = B$: left-multiply by A^{-1}, *keeping the order*: $X = A^{-1}B$ (never BA^{-1}).

Q4 — the real one

$AX(D + BX)^{-1} = C$: right-multiply by $(D + BX)$: $AX = C(D + BX) = CD + CBX$; gather X on the *right*: $(A - CB)X = CD$, so

$$X = (A - CB)^{-1}CD.$$

Q6 — a false shortcut

"$(A + B)^{-1} = A^{-1} + B^{-1}$": **False**. Counterexample (diagonals): $A = 2I$, $B = 3I$:

$$(A + B)^{-1} = \tfrac{1}{5}I \neq \tfrac{5}{6}I = A^{-1} + B^{-1}.$$

The two order rules

$(AB)^{-1} = B^{-1}A^{-1}$ reverses; and X factors out on the side it sits: $(A - CB)X = CD$, not $X(A - CB) = CD$.

D · The Rank Theorem, and Q8 step by step (Q7, Q8)

Rank Theorem: $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$ (number of columns)

- **Q7:** A is 10×9 with rank 7: nullity $= 9 - 7 = \mathbf{2}$.

Step 1 — RREF of the 4×4 matrix

$$R = \begin{bmatrix} 1 & 0 & 0 & 1/2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -13/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 pivots; column 4 free $\Rightarrow$ nullity $= 1$.

Step 2 — substitute the free variable

$$\begin{aligned} x_1 + \tfrac{1}{2}x_4 = 0 &\quad \Rightarrow x_1 = -\tfrac{1}{2}x_4 \\ x_2 = 0 &\quad \Rightarrow x_2 = 0 \\ x_3 - \tfrac{13}{2}x_4 = 0 &\quad \Rightarrow x_3 = \tfrac{13}{2}x_4 \end{aligned}$$

Setting the free variable $x_4 = 1$: $\mathbf{v} = \left(-\tfrac{1}{2},\ 0,\ \tfrac{13}{2},\ 1\right)^\top$.

Step 3 — substitute $\mathbf{v}$ back into the ORIGINAL A (row by row)

$$\begin{aligned} &\text{row 1: } 2(-\tfrac{1}{2}) - 0 + 0 + 1 = 0 && \checkmark \\ &\text{row 2: } -\tfrac{5}{2} + \tfrac{13}{2} - 4 = 0 && \checkmark \\ &\text{row 3: } \tfrac{1}{2} - \tfrac{13}{2} + 6 = 0 && \checkmark \\ &\text{row 4: } -8(-\tfrac{1}{2}) - 0 - 2(\tfrac{13}{2}) + 9 = 0 && \checkmark \end{aligned}$$

all null vectors $= \operatorname{span}\{\mathbf{v}\}$.

D · Q8 and Q9: rank and nullity

The results (Q8's full derivation: previous slide):

	Result after elimination	rank	nullity $= n -$ rank
Q8 (4×4)	3 pivots; RREF has one zero row	3	$4 - 3 = \mathbf{1}$ (null vector $(-\frac{1}{2}, 0, \frac{13}{2}, 1)^\top$)
Q9 (6×4)	3 pivots; three zero rows	3	$4 - 3 = \mathbf{1}$ (null vector $(46, -5, -103, 37)^\top$)

- Both null vectors were verified: $A\mathbf{v} = \mathbf{0}$ exactly. Q9 is tall (6×4) — rank at most 4, and its rows are rank-3 dependent: the extra rows are combinations.

E · Q10 in detail: the two substitutions

Step 1 — $L\mathbf{y} = \mathbf{b}$ (forward)

$$\left[\begin{array}{cc|c} 1 & 0 & -13 \\ -1 & 1 & 9 \end{array}\right]$$

row 1: $y_1 = -13$

row 2: $-y_1 + y_2 = 9$

$\Rightarrow y_2 = 9 + y_1 = 9 - 13 = -4$

$\mathbf{y} = (-13, -4)^\top$

Step 2 — $U\mathbf{x} = \mathbf{y}$ (backward)

$$\left[\begin{array}{cc|c} 1 & 9 & -13 \\ 0 & 2 & -4 \end{array}\right]$$

row 2: $2x_2 = -4 \quad \Rightarrow x_2 = -2$

row 1: $x_1 + 9x_2 = -13 \quad \Rightarrow x_1 = -13 - 9(-2) = 5$

$\mathbf{x} = (5, -2)^\top$

Step 3 — substitute $\mathbf{x}$ back into the ORIGINAL $A = LU = \left[\begin{smallmatrix} 1 & 9 \\ -1 & -7 \end{smallmatrix}\right]$

row 1: $1 \cdot 5 + 9 \cdot (-2) = 5 - 18 = -13$ ✓

row 2: $-1 \cdot 5 + (-7)(-2) = -5 + 14 = 9$ ✓

$A\mathbf{x} = \mathbf{b}$ ✓

E · Solving with LU: two easy triangular steps (Q10)

Given $A = LU = \left[\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 9 \\ 0 & 2 \end{smallmatrix}\right]$, solve $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} = (-13, 9)^\top$

Step 1 — $L\mathbf{y} = \mathbf{b}$ (forward): $y_1 = -13$; $-y_1 + y_2 = 9 \Rightarrow y_2 = 9 + (-13) = -4$.
Step 2 — $U\mathbf{x} = \mathbf{y}$ (backward): $2x_2 = -4 \Rightarrow x_2 = -2$;
$x_1 + 9x_2 = -13 \Rightarrow x_1 = -13 + 18 = 5$.

$$\mathbf{x} = (5, -2)^\top, \qquad \underbrace{\left[\begin{smallmatrix} 1 & 9 \\ -1 & -7 \end{smallmatrix}\right]}_{A=LU}\left[\begin{smallmatrix} 5 \\ -2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -13 \\ 9 \end{smallmatrix}\right] \ \checkmark$$

- Two triangular sweeps replace one full elimination — that is the whole point of precomputing LU.

E · Building LU: record the multipliers (Q11)

Eliminate A (4×3) with row additions only; each multiplier is stored in L:

$$\begin{array}{llll} R_2 \leftarrow R_2 - 2R_1 & (L_{21} = 2) & R_3 \leftarrow R_3 - 3R_2 & (L_{32} = 3) \\ R_3 \leftarrow R_3 + 3R_1 & (L_{31} = -3) & R_4 \leftarrow R_4 + R_2 & (L_{42} = -1) \\ R_4 \leftarrow R_4 - 0R_1 & (L_{41} = 0) & R_4 \leftarrow R_4 - 2R_3 & (L_{43} = 2) \end{array}$$

$$U = \begin{bmatrix} 1 & -3 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -3 & 3 & 1 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix}, \qquad LU = A \checkmark$$

- U's last row is zero (no 4th pivot) — and L is 4×4 while U is 4×3: sizes differ (that is Q14 **False**).

E · Q11 reasoning: why L and U take this shape

Elimination = elementary matrices

Each addition $R_i \leftarrow R_i - mR_c$ is $E_i A$ with $E_i = I$ plus $-m$ at position (i, c). Stacking them:

$$E_4 E_3 E_2 E_1 A = U \implies A = \underbrace{(E_1^{-1} \cdots E_4^{-1})}_{=L} U.$$

Why L is unit lower triangular

E^{-1} of a row addition is the same addition with $+m$ at (i, c) — below the diagonal, ones on it. Products of such matrices stay unit lower triangular, and the multipliers m we stored during elimination appear unchanged in L.

Why U's last row is zero

A is 4×3: at most 3 pivots. Row 4 eliminated to $\mathbf{0}$ because it was already a combination of the others — L's row 4 holds the coefficients that rebuild A's row 4 from the *rows of* U:

$$\text{row}_4(A) = 0 \cdot \text{row}_1(U) + (-1) \cdot \text{row}_2(U) + 2 \cdot \text{row}_3(U)$$

$$= [0, -2, -2] \checkmark$$

After elimination only zeros remain in U's row 4, so U ends in a zero row and $\text{rank}(A) = 3$.

Why no swaps were needed

Every pivot was nonzero when reached (1, then 2, then -1). With a zero pivot we would need a row interchange $PA = LU$ instead.

E · Inverting via LU: reverse the order (Q12, Q13, Q14)

Q12 — $A^{-1} = U^{-1}L^{-1}$, one factor at a time

$L^{-1} = \left[\begin{smallmatrix} 1 & 0 \\ 6 & 1 \end{smallmatrix}\right]$ (flip sign of -6); $U^{-1} = \begin{bmatrix} -1/3 & -5/7 \\ 0 & 1/7 \end{bmatrix}$;

$$A^{-1} = U^{-1}L^{-1} = \left[\begin{smallmatrix} -1/3 & -5/7 \\ 0 & 1/7 \end{smallmatrix}\right] \left[\begin{smallmatrix} 1 & 0 \\ 6 & 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -97/21 & -5/7 \\ 6/7 & 1/7 \end{smallmatrix}\right]$$

Check: $AA^{-1} = I$ ✓ (note: $L^{-1}U^{-1}$ would be wrong.)

Q13 — L diagonal

Pick A already in echelon form, $L = I_3$:
$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 0 & 0 \end{bmatrix} = I_3 A$.

Q14 — same dimensions?

False: L is $m \times m$, U is $m \times n$ — Q11 gave 4×4 vs. 4×3.

F · A row operation as a matrix (Q17)

Build E by performing $R_4 + 3R_1 \to R_4$ on the identity

$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix}$ — the 3 sits in position $(4, 1)$: row 4 will collect $3\times$row 1.

Test: EA performs the operation

$A = \begin{bmatrix} 1 & -5 \\ 5 & -3 \\ 1 & -4 \\ 3 & -2 \end{bmatrix}$: rows 1–3 pass through, row 4 = row 4 + 3 row 1

$= [3, -2] + [3, -15] = [6, -17]$:

$$EA = \begin{bmatrix} 1 & -5 \\ 5 & -3 \\ 1 & -4 \\ 6 & -17 \end{bmatrix} \checkmark$$

- Left-multiplying by an elementary matrix = performing its row operation — the bridge between row reduction and matrix algebra, and the reason $[\,A \mid I\,] \sim [\,I \mid A^{-1}\,]$ works.

One table: the whole assignment

Idea	The one thing to remember
Products	inner dimensions match; $AB \neq BA$ (Q1); $A^\top A$ is 1×1 (Q15); shapes decide definedness (Q16)
Inverses	$\det \neq 0$ + the 2×2 formula (Q2, Q3); $(A+B)^{-1}$ is *not* $A^{-1}+B^{-1}$ (Q6)
Matrix equations	left-solve $AX=B$; for $AX(D+BX)^{-1}=C$: gather X on the right, $X=(A-CB)^{-1}CD$ (Q4, Q5)
Rank & nullity	$\text{rank}+\dim\text{Nul}=n$ (Q7: $9-7=2$; Q8, Q9: rank 3, nullity 1)
LU	additions build L, elimination builds U (Q11); solve in two triangular sweeps (Q10); $A^{-1}=U^{-1}L^{-1}$ reversed (Q12); sizes $m\times m$ vs $m\times n$ (Q13, Q14)
Elementary matrices	row op on I; EA performs it (Q17)

Review map — back to the class slides

Questions	Topic to review
Q1 Q15 Q16	matrix operations, multiplication law, transposes
Q2 Q3 Q6 Q14	2×2 inverse formula; inverses of products; elementary matrices
Q4 Q5	algebraic manipulation with inverses
Q17	elementary matrices as row operations
Q7 Q8 Q9	nullity, dimension, rank, Rank Theorem
Q10–Q13	LU factorization and its uses

Answer key (instructor copy, reference only)

- Q1: $AB = [[-1, -5], [14, 5]]$; $BA = [[5, 5], [-14, -1]]$; False
- Q2: yes ($\det = -1$); $M^{-1} = [[-4, 3], [3, -2]]$
- Q3: $c \neq 0$ and $c \neq 1$
- Q4: $X = (A - CB)^{-1}CD$
- Q5: $X = A^{-1}B$
- Q6: False
- Q7: nullity $= 2$
- Q8: rank 3, nullity 1
- Q9: rank 3, nullity 1
- Q10: $\mathbf{x} = (5, -2)^{\top}$
- Q11: $L =$ $[[1, 0, 0, 0], [2, 1, 0, 0], [-3, 3, 1, 0], [0, -1, 2, 1]]$; $U = [[1, -3, 2], [0, 2, 0], [0, 0, -1], [0, 0, 0]]$
- Q12: $A^{-1} = [[-97/21, -5/7], [6/7, 1/7]]$
- Q13: e.g. $A = [[1, 2], [0, 3], [0, 0]]$
- Q14: False
- Q15: $A^{\top}A = 29$; $AA^{\top} = [[4, -6, -8], [-6, 9, 12], [-8, 12, 16]]$
- Q16: A, D, E, F
- Q17: $E = [[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [3, 0, 0, 1]]$; $EA = [[1, -5], [5, -3], [1, -4], [6, -17]]$

Thank you!

Questions? Comments? Corrections?

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